A Mathematic Framework for Multimedia Content Retrieval and Summarization

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Abstract **–** The goal of both the upcoming MPEG-7 standard and the recent Semantic Web initiative is to provide standardized ways for metadata description to maximize the interoperability of the multimedia content and facilitate the creation of novel multimedia applications. A close examination reveals that the basic data representation accepted by both the MPEG-7 standard and the Semantic Web is a graph of nodes and arcs. We then mathematically formulate the problems of information retrieval and summarization in the context of the MPEG-7 standard and the Semantic Web as graph retrieval and summarization. Algorithms are presented for graph retrieval and summarization. We discuss the information required by the proposed mathematic framework as well as how to support the description of information in the context of the MPEG-7 standard and the Semantic Web.

Index Terms – MPEG-7, the Semantic Web, nodes and arcs, multimedia content retrieval and summarization.

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I. INTRODUCTION

The explosive growth of digital multimedia content makes the selection and searching of multimedia information harder and harder. It is well known the performances of currently available multimedia information search engines are far from satisfying users' needs. Instead of pushing the performances of the search algorithms alone, the Moving Picture Experts Group (MPEG) of the International Standards Organization initiated MPEG-7 to help finding better solutions for multimedia-centric search by improving the descriptions of multimedia content. The goal of MPEG-7 standard [12] is to allow interoperable retrieval, browsing, filtering, and summarization of multimedia content. MPEG-7 Description Schemes (DSs) [11], [17] are metadata structures and provide a standard way for describing, annotating, ma\naging, and accessing multimedia content in the MPEG-7 Description Definition Language (DDL). The MPEG-7 DDL is based on the eXtended Markup Language (XML) [2]. XML is accepted as the emerging standard for data interchange on the Web and allows users to add arbitrary structure to the data.

At the same time, realizing that it is very difficult to either automate everything on the World Wide Web or manually manage the information on the Web, researchers proposed the Semantic Web as an extension of the current Web [1]. The Semantic Web tries to bring structure to the machine-readable content of Web pages to make the information on Web pages machine-understandable. Hence, computer software agents can process the semantics of the content on Web pages and carry out sophisticated tasks (e.g., retrieve information, summarize information, manipulate online resources, etc.) on the Internet for users.

Both the MPEG-7 standard and the Semantic Web try to make the information computable by computers. They adopt that same approach: Making the data machine-understandable via the cooperation between computers and users. This unavoidably leads to explore and annotate the structure of the data – the basic of computer algorithms. They have their own emphases and are complimentary to each other to some degree.

The MPEG-7 DSs (e.g. Segment DS, AudioSegment DS, VideoSegment DS, Semantic DS, Event DS, etc.) spans a wide range of abstraction levels: the low-level, the intermediate levels, and the high-level. The low-level description describes the perceptual features of the content. For example, the MPEG-7 visual standard for content description is being developed to specify images and video so that the similarities in images or video can be measured based on visual criteria [18]. An image can be described as a group of image segments which are characterized by the segments' shape, size, texture, color, and so on. The high-level descriptions (e.g. Semantic DS, Object DS, etc.) provide conceptual information of the data understandable by the human beings. The intermediate level descriptions fill the gap between the low-level descriptions and the high-level description and can provide models that link the perceptual features with the semantic concepts.

The challenge of the Semantic Web is to define a language that enables both computers and users to express both data and

rules for inferring about the data. XML is accepted by the Semantic Web to provide a standardized way for users to describe their documents structurally. XML enables users to define the syntax structure of the data. However, it does not provide ways to define the meaning of the structures [1], which is required by reasoning. Other models and model instances, e.g., Resource Description Framework (RDF) [10] and Resource Description Framework Schemas (RDFSs) [14], have been proposed to add logic to the structures of Web documents so that the information can be inferred. The basic information elements concerned by the Semantic Web are the semantic concepts described by text, which are the main concerns of the high-level description schemes of the MPEG-7 MDSs.

The MPEG-7 MDS and RDF can be used jointly to provide richer descriptions for multimedia content on the Internet. The user can use MPEG-7 MDS to describe detailed information, which include low-level (appearance level), intermediate-level, and high-level (semantic level) information about multimedia resources (e.g., images, audio, video, etc.), and use RDF to facilitate reasoning about the semantic information. Two multimedia resources may have dramatically different low-level information but the same semantic meaning. If only the low-level information is described, the user will have difficulties to tell that those two resources represent the same semantic concept. If only the semantic information is described, the differences between those two resources, which share the same semantic meaning, are neglected. Moreover, semantic concepts have dynamical aspects. A multimedia resource may represent different semantic concepts in different reasoning contexts. Although the MPEG-7 standard defines DSs for description of the semantic of content, those DSs are not particularly designed for reasoning purpose. Depending on applications, users may emphasize on different combinations of the low-level information, the intermediate-level information, the semantic information, and the semantic reasoning. The combination of the MPEG-7 MSD and RDF will facilitate the creation of novel and full-scale multimedia applications.

One of the main goals of describing multimedia content is for multimedia data searching, which is based on data comparison and makes data retrieval and summarization possible. It is very important to consider machine-computable issue while finalizing the MPEG-7 standard and developing the Semantic Web. Given current status of the MPEG-7 standard and the Semantic Web, this paper presents algorithms and techniques for data comparison, searching, and summarization. We begin with discussing information representation which lays down the base for information computation in Section II. In the context of the MPEG-7 standard, we are particularly interested in DSs for content description, which are addressed as Content DSs (CDSs) in the rest of the paper. The algorithms for comparing, matching, and summarizing data are presented in III. Experimental results are presented in Section IV. Finally, the paper finishes with conclusions in Section V.

II. UNIFIED INFORMATION REPRESENTATION

Despite of the differences between the emphases of RDF and those of the MPEG-7 standard, they both use XML which follows Backus-Naur Form notation (BNF). Therefore, the information items described by either the MPEG-7 CDSs or RDF can be converted into graphs that consist of nodes and arcs.

A. Information Representation of the MPEG-7 CDS.

In the MPEG-7 standard, the nodes can represent image segments, objects, images, video segments, audio segments, and so on. The properties of the nodes can be color histograms of image segments, texture histograms of image segments, textual annotations of images or video segments, temporal durations of video segments, and so on. The relations among the nodes can be the spatial relations among image regions in an image, the temporal relations among segments in multimedia stream sequence, the ownership (e.g., is composed of) between an image sequence and the images/objects in the sequence, and so on. The properties of the relations describe the quantitative or analytic values of the relations. Figure 1 shows an example of using the MEPG-7 Segment DS and SegmentRelation DS to describe an image.

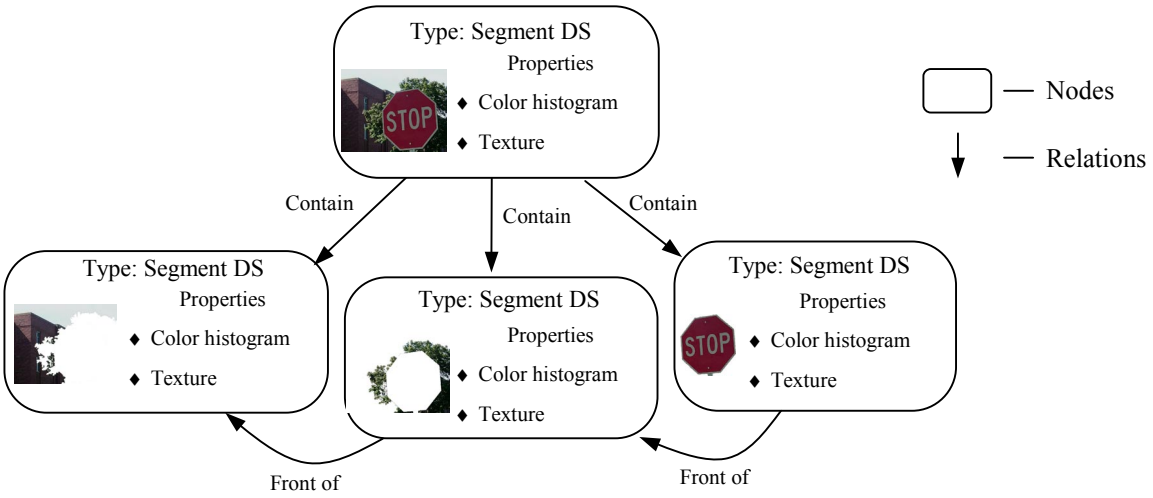


Figure 1. An example of image description using the MPEG-7 CDS. The images in the nodes are for visualization purpose only.

B. Information Representation of RDF

In the context of the Semantic Web, the user can use RDF data models to describe a set of related resources as a so-called entity-relationship diagram (ER diagram). An ER diagram is an RDFS – an instance of a RDF data model. The EM diagram consists of nodes and directional arcs. A node of an EM diagram represents an entity that can be an XML element within a Web page, a Web page, or a collection of Web pages. Arcs of an EM diagram represent the relations among those entities. The relations are associated with named properties. Examples of the properties of the relations can be “has address”, “is the creator of”, and so on. Two nodes connected by an arc denotes an RDF statement and makes assertions that one particular thing (e.g. people,

Web page, etc.) represented by one of the two nodes has properties (e.g. "has homepage", "is the creator of", etc.) with certain value (e.g., Web page, people, etc.) represented by the other node. Consider a simple example:

Example I: *The creator of the http://www.ifp.uiuc.edu/~hong is a person whose name is Pengyu Hong and has an email address as hong@ifp.uiuc.edu and a picture at http://www.ifp.uiuc.edu.edu/~hong/pic.jpg*.

Both "*http://www.ifp.uiuc.edu/~hong*" and "*http://www.ifp.uiuc.edu/~hong/pic.jpg*" are resources. The creator is an instance of data type "person" that has name, email address and picture. The name "*Pengyu Hong*" and the email address "*hong@ifp.uiuc.edu*" are of string literals. A literal is the smallest element that cannot be further evaluated by the RDF processor and may have content that is XML markup. Figure 2 shows the EM diagram of Example I.

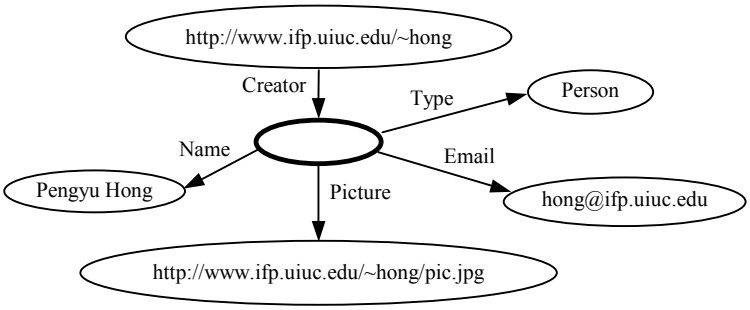


Figure 2. The ER diagram of Example I. Only the data type of the creator is explicitly shown.

C. Combine the MPEG-7 MDS and RDF

We can jointly use the MPEG-7 CDS and RDF to describe multimedia resources on the Internet. The MPEG-7 standard is used to enrich the description of the multimedia semantic units. RDF is used to describe relationships among the semantic units. Figure 3 shows an example of combining the MPEG-7 CDS and RDF to describe the Example I.

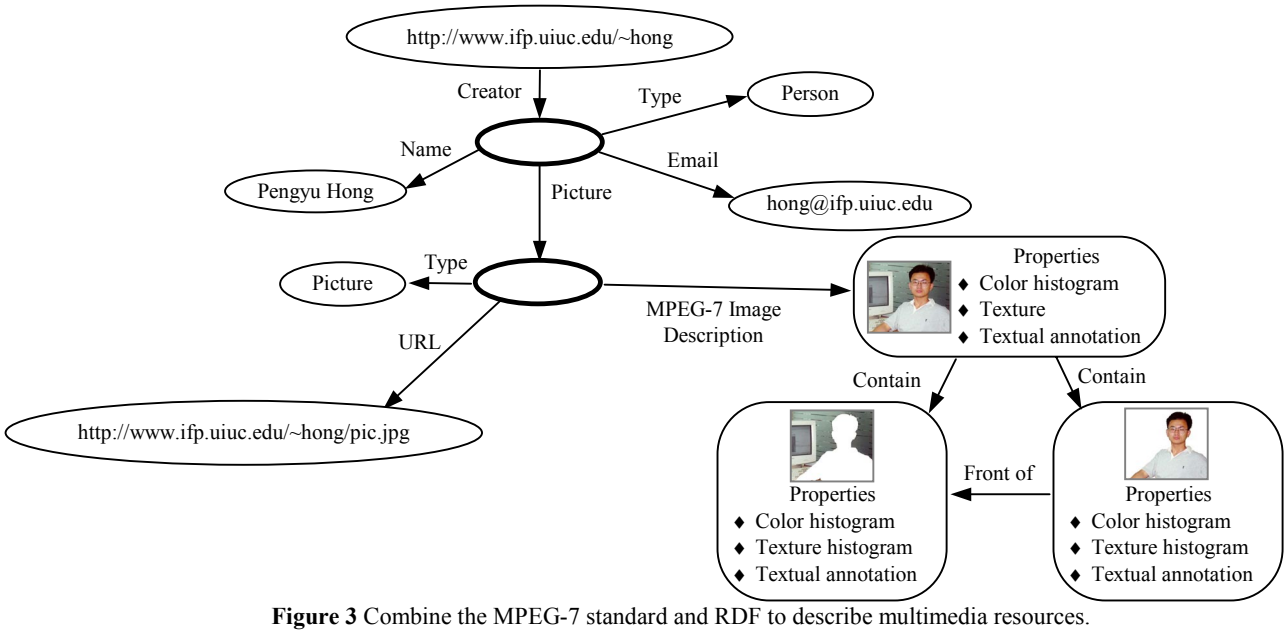


Figure 3 Combine the MPEG-7 standard and RDF to describe multimedia resources.

D. Unify the Representation

Although both the MPEG-7 and RDF use XML to describe information, their graph representations are not exactly the same due different emphases. Small modifications should be done to unify the representation so that a unified mathematical framework can be proposed to compute multimedia information. RDF and the MEPG-7 standard share the same primitive data types defined by XML. Based on those primitive data types, other structural data types are hierarchically defined. We will use attributed relational graph (ARG) [20] as the unified representation. An ARG is a graph consists of nodes and arcs. A node represents an information unit and is associated with an attribute vector describing the properties of the node. An arc represents the relation between two nodes connected by the arc. An arc also has a feature vector describing the value of the relation.

A MPEG-7 CDS graph is similar to ARG consisting of heterogeneous nodes (e.g. Segment DS, Semantic DS, AudioSegment DS, etc.). We can convert a MPEG-7 CDS graph to an ARG by using one attribute vector to store the information represented by the MPEG-7 Descriptors of a node and adding a data type ID to denote the description type of that node.

In an RDF graph, information is represented by units that are as fine as possible. A node represents either a resource specified by a Uniform Resource Identifier (URI) or a literal of a primitive data type defined by XML. An RDF graph even uses another node to denote the data type of its parent node. We can convert an RDF graph to an ARG such that: (1) No node in the graph is used to represents the data type of the node; (2) No node is of a primitive data type. This can be done by moving the information stored in the above two kinds of nodes to their parent nodes. We use an intrinsic (or build-in) string literal in the node to represent the data type of the node. Basically, the string literal records the name of the data type. For a node of a primitive data type, say node *c*, we add a corresponding property item in its parent indicated by node *p*. The name of the property item is the named property represented by the relation connecting *c* and *p*. The value of the property item is the information stored at *c*. Following the above method, we can convert the graph in Figure 2 into two graphs in Figure 4.

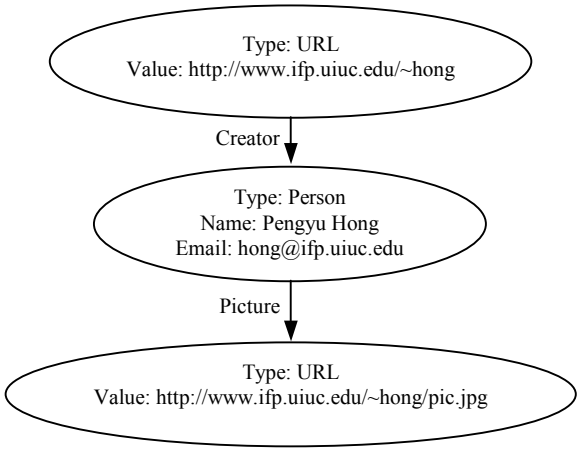


Figure 4. Convert an RDF graph into its corresponding attributed relational graph.

E. Mathematical Abstract of the Unified Graph Representation

Let a graph $G_i = < A_i, R_i >$.

(a) $A_i = \{\langle o_{ik}, \vec{a}_{ik} \rangle\}_{k=1}^{U_i}$ denotes the node set of G_i. U_i is the number of the nodes. o_{ik} is a node in G_i. $\vec{a}_{ik}$ is the attribute vector of o_{ik}. $\vec{a}_{ik}$ may contain heterogeneous information (e.g. image features, text string, audio features, etc.). The heterogeneous information can be represented by separate vectors. Nonetheless, this kind of description is convenient for description. When it comes to implemental issues, extra attention should be paid to handle the heterogeneous information.

(b) $R_i = \{\langle r_{icd}, \vec{b}_{icd} \rangle\}_{c,d=1}^{U_i}$ denotes the relations among the nodes. r_{icd} is a relation connecting two nodes o_{ic} and o_{id}. $\vec{b}_{icd}$ is the attribute vector of r_{icd} . $\vec{b}_{icd}$ can represent binary relations (e.g., "is the creator of", "is mother of", etc.) or continuous relations (e.g. relative spatial distance, relative size, etc.). Similar to $\vec{a}_{ik}$, $\vec{b}_{icd}$ may contain information of heterogeneous types. Self-relation, which describes the relation between a node and itself, is allowed. We have $\vec{b}_{icd} = \vec{b}_{idc}$ if the relations are directionless.

III. Information Retrieval and Summarization as Graph Matching and Summarization

Last section shows that a multimedia information item that spans one or more description levels (i.e. the low-level, intermediate-level, semantic-level, and reasoning-level) can be represented as an ARG. Based on this unified representation, this section points out that information retrieval problem is equal to graph matching problem. We further present a framework for learning a compact pattern ARG (PARG model) to summarize multiple information items. The unified representation allows us to use the approaches presented in this section to retrieve or summarize information regardless the number of description levels spanned by information items.

A. Information Retrieval as Graph Matching

To retrieve information items, we need to measure the similarity between query and each information item in the database. Since information items can be represented as ARGs, the basic problem of information retrieval becomes measuring the similarity between two graphs. Therefore, we can utilize two-graph matching techniques.

Two-graph matching is a fundamental problem and has been widely investigate [3], [8], [16], [21], [23]. The goal of two-graph matching is to match the nodes of one graph with the nodes of the other graph. It is necessary to realize that the matching results are not exact ones because the information encoded in the nodes and relations of graphs has variance. For example, the low-level

feature of two images of one object may not be the same. At the semantic level, variances may be introduced by different understanding on the same multimedia content of different creators or the naming schemes of information due to the existence of homonyms and synonyms. A solution to this problem is to use ontologies [22], which are not the focus of this paper. It is also necessary to realize that information units may seem to have the same meaning, however differ from each other due to their contexts. Fortunately, the context information is considered by graph matching algorithms.

To match two graphs, we need to measure the similarities between each pair of nodes and those between each pair of relations. The similarities should be measured based on the attribute vectors of nodes and relations. The exact way to measure those similarities should be application dependent. Without losing generality, we will assume that measurement methods are already available. We then adopt the approach proposed in [3] to calculate the matching from the graph of an information item in database to the graph of query.

Let $G_q = < A_q, R_q > = \langle o_{q0}, \{\langle o_{qk}, \vec{a}_{qk} \rangle\}_{k=1}^{U_q}, \{\langle r_{qcd}, \vec{b}_{qcd} \rangle\}_{c,d=1}^{U_q} \rangle$ denote the graph of query. o_{q0} is a null node which does not have physical existence. It is added to the query graph per the request of the matching algorithm. It provides a matching destination for those nodes that are in G_i and are not related to the information represented by G_q. Let G_i denote the graph of an information item. The algorithm calculates the probability of $o_{ik} \in G_i$ matching to $o_{qm} \in G_q$ as a conditional probability $P(o_{ik} = o_{qm} \mid G_q, G_i)$. Let $\boldsymbol{M}_q(o_{ik}) = \arg\max_{o_{qn}} P(o_{ik} = o_{qn} \mid G_q, G_i)$. We then define the degree that G_i is similar to G_q as

$$S(G_i, G_q) = \sum_{o_{ik}} P(q_{ik} = M_q(o_{ik}) \mid G_q, G_i) \aleph(M_q(o_{ik})) \tag{1}$$

where

$$\aleph(M_q(o_{ik})) = \begin{cases} 1 & \text{if } M_q(o_{ik}) \neq o_{q0} \\ 0 & \text{if } M_q(o_{ik}) = o_{q0} \end{cases}$$

B. Information Summarization as Graph Summarization

1) Problem formulation

Since the information items are represented as ARGs, the information calculation is based on ARGs. Hence, summarizing multiple information items in the database is equal to summarizing multiple ARGs. Given the above similarity measurement method for information items, this section will represent a framework for building a compact PARG model to summarize a large set of sample ARGs (SARGs) representing information items. The PARG model encodes the characteristics of both the spatial properties and non-spatial properties of the instances of the pattern. The PARG model can be used to detect the instance of the pattern

later. We use $\boldsymbol{P}$ to denote the PARG model and use $\boldsymbol{G} = \{G_i\}_{i=1}^{T}$ to denote the SARG set. Each G_i represents a data item (e.g., an image). We assume that the $\boldsymbol{G}$ is governed by some underlying probabilistic distribution function (PDF) $f(G)$, where G is a variable represents a graph drawn from $\boldsymbol{G}$. Our goal is to estimate $f(G)$ given $\boldsymbol{G}$. This is often treated as a statistical inference problem that estimates a probabilistic model $p(G)$ to approximate $f(g)$ by minimizing the Kullback-Leibler divergence $KL(f(G)\| p(G))$ [4]:

$$p(G)^* = \arg\min_{p(G)} KL(f(G) \| p(G)) = \arg\min_{p(G)} \int f(G) \log \frac{f(G)}{p(G)} dG = \arg\min_{p(G)} E_{f(G)}[\log f(G)] - E_{f(G)}[\log p(G)] \quad (2)$$

The PARG model should encode the full information of $p(G)$. Since the first term in eq. (1) is a constant regardless what $f(G)$ exactly is, the problem of eq. (2) is equal to

$$p(G)^* = \arg\max_{p(G)} E_{f(G)}[\log p(G)] \quad (3)$$

In practice, we only have limited samples. Therefore, we approximate the problem of eq. (3) by

$$p(G)^* \approx \arg\max_{p(G)} \sum_{G_i} f(G_i) \log p(G_i) \quad (4)$$

It is in general difficult to directly estimate $p(G)$ without any prior knowledge about $f(G)$. In practice, $f(G)$ is usually assumed to be a parametric function whose parameter set is denoted by $\widetilde{\Theta}$. Consequentially, the PARG model is a parametric model whose parameter set is $\widetilde{\Theta}$. We can rewrite eq. (4) as

$$\widetilde{\Theta}^* \approx \arg\max_{\widetilde{\Theta}} \sum_{G_i} f(G_i \mid \widetilde{\Theta}) \log p(G_i \mid \widetilde{\Theta}) \quad (5)$$

Since the form of a function can be very complicated, we further assume $f(G)$ is a linear combination of a set of mixture density functions to make the problem tractable. Therefore, we design the PARG $\boldsymbol{\Gamma}$ to consist of a small set of model components. Each model component of $\boldsymbol{\Gamma}$ is a parametric ARG and corresponds to a mixture of $f(G)$. Each node in the PARG is associated with an attributed PDF described by the parameters of that node. Each relation in the PARG model is associated with a relational PDF described by the parameters of that relation. Therefore, $p(G)$ consists of a set of attributed PDFs and a set of relational PDFs. We further assume that the forms of the attributed PDFs and the relational PDFs are Gaussian

$$p(\vec{x}) = \frac{\exp[-\frac{1}{2}(\vec{x} - \vec{u})^T C^{-1} (\vec{x} - \vec{u})]}{(2\pi)^{\xi/2} \mid C \mid} \quad (6)$$

where $\vec{x}$ is either an attribute vector, $\vec{u}$ is the mean of the Gaussian distribution, C is the covariance matrix of the Gaussian distribution, and ξ is the dimension of $\vec{x}$. $\vec{x}$ may contain heterogeneous information (e.g. color, texture, text, etc.). Different kind of information may not be compared with each other. Hence, the covariance matrix C may be a block diagonal matrix. Each

matrix block represents a covariance sub-matrix for one kind of information or multiple kinds of comparable information.

2) Notations for PARG

We have defined the notation for SARG in Section II.E. We now define the following mathematic notations for the PARG model. The PARG model $\boldsymbol{\Gamma}$ has M components: $\boldsymbol{\Gamma} = \{\Phi_w\}_{w=1}^{M} = \left\{\left\langle \Omega_w, \Psi_w, \Theta_w \right\rangle\right\}_{w=1}^{M}$. Ω_w is the model node set. Ψ_w is the model relation set. Θ_w is the parameter set.

- $\Omega_w = \{\omega_{wk}\}_{k=0}^{N}$. ω_{w0} is null model node. ω_{wk} (k>0) is a non-null model node. ω_{w0} does not have physical existence. It is used to provide a modeling destination of those sample nodes that are not instances of the model nodes. N is the number of the model nodes. Actually, different model components may have different number of valid model nodes. N should be the largest number of the non-null model nodes in all the components. If a component has less than N valid nodes, null nodes that do not have physical instances can be added. The reason that we assume all components have same number of nodes is to make problem description easier. However, it won't affect the modeling capability of the PARG model.
- $\Psi_w = \{\psi_{wcd}\}_{c,d=1}^{N}$. Each sample relation ψ_{wcd} model the relation between ω_{wc} and ω_{wd}.
- $\Theta_w = \left\langle \alpha_w, \{e_{wk}\}_{k=1}^{N}, \{\beta_{wk}\}_{k=1}^{N}, \{f_{wcd}\}_{c,d=1}^{N} \right\rangle$ is the parameter set of Φ_w. The whole parameter set of $\boldsymbol{\Gamma}$ is $\widetilde{\Theta} = \{\Theta_w\}_{w=1}^{M}$ The value of α_w denotes the percentage of the pattern instances in $\boldsymbol{G}$ is represented by Φ_w. The value of $\alpha_w S$ implies the number of the pattern instances in $\boldsymbol{G}$. $e_{wk} = \left\langle \vec{\mu}_{wk}, \Sigma_{wk} \right\rangle$ denotes the parameters of the attributed PDF represented by the model node ω_{wk}. $\vec{\mu}_{wk}$ and Σ_{wk} are the mean and covariance matrix of the attributed PDF respectively. β_{wk} denotes the percentage of samples nodes that are represented by ω_{wk} and belong to the instances represented by Φ_w. $f_{wcd} = \left\langle \vec{\gamma}_{wcd}, \Lambda_{wcd} \right\rangle$ denotes the parameters of the relational PDF represented by the model relation ψ_{wcd}. $\vec{\gamma}_{wcd}$ and Λ_{wcd} are the mean and covariance matrix of the relational PDF respectively.

3) Illustration of the problem formulation

Using the above problem formulated and the notations, we can an example to illustrate how we formulate the problem (see Figure 5). The number of model components M should be much smaller so that of samples T so that we have a compact representation of the pattern.

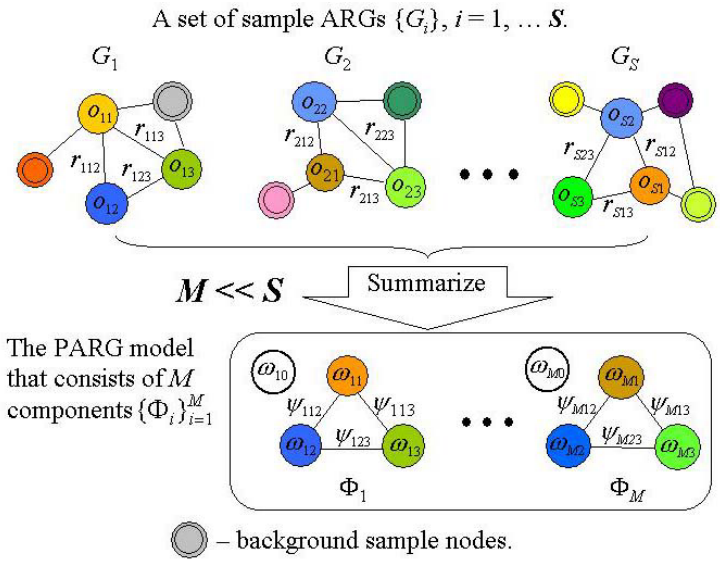


Figure 5. Illustration of the problem formulation of Graph Summarization.

C. *Infer the Parameters of the PARG model*

Given the above problem formulation, we proposed to estimate both the structure and parameters of the PARG model $\boldsymbol{\Gamma}$ iteratively (see Figure 6). The procedure is explained as following.

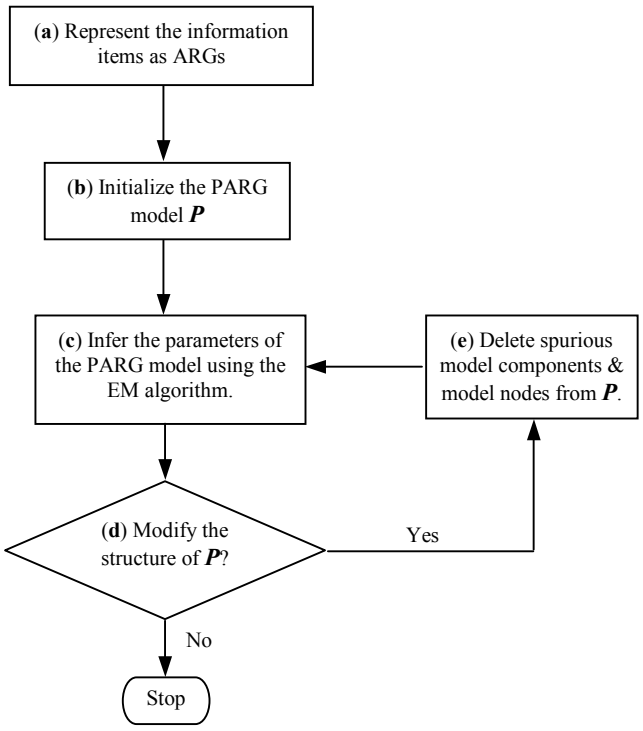


Figure 6. The learning procedure of the PARG model.

1) Step (a)

Represent each data sample as an ARG. Take images as an example. We can represent each pixel of an image as a node in an ARG and the relations (e.g., distance) among the pixels as the arcs in that ARG. Or, we can apply image segmentation or edge detection to the images. The image segments or detected edges are represented as nodes in the ARGs. The relations of the ARGs represent the spatial relations among those image segments or edges.

2) Step (b)

We initialize the PARG model $\boldsymbol{\Gamma}$ by the following algorithm so that the components of $\boldsymbol{\Gamma}$ are as different as possible.

Algorithm I: Initialize the PARG Model $\boldsymbol{\Gamma}$

- Randomly select one SARG G_i from $\boldsymbol{G}$ and use G_i to initialize Φ_1. First, we assign the structure of G_i to Φ_1. The attribute vectors and relation vectors of G_i are used to initialize the corresponding attribute means of the attributed PDFs and the relation means of the relational PDFs. The attributed covariance matrixes and the relational covariance matrixes are initialized as identical matrixes.
- ***for*** $k = 2$ ***to*** M
  - Calculate the matching score $S(G_j, \Phi_m)$ from G_j to each Φ_m $(1 \le m < k)$ using eq. (1).
  - Select an SARG $G_x = \arg\min_{G_j} \sum_{m=1}^{k-1} S(G_j, \Phi_m)$.
  - Use the properties of G_x to initialize Θ_k.
- ***end-for***

3) Step (c)

This step estimates the parameters of $\boldsymbol{\Gamma}$ via the Expectation-Maximization algorithm [5]. The EM algorithm runs iteratively between two steps, the Expectation step (E-step) and the Maximization step (M-step), until it converges or reaches the maximum number of iterations.

Algorithm II: Estimate the parameters of $\boldsymbol{\Gamma}$ via the Expectation-Maximization algorithm

- **E-step:** Calculate the matching probabilities between each SARG and the PARG model $\boldsymbol{\Gamma}$.
- **M-step:** Use the calculated matching probabilities to update the parameters of $\boldsymbol{\Gamma}$.

In [9], we proved that the parameters of the PARG model should be updated using the following expression in the M-step:

$$P(G_i = \Phi_w \mid \widetilde{\Theta}^{(t)}) = \frac{S(G_i, \Phi_w)}{\sum_{m=1}^{M} S(G_i, \Phi_m)}$$

$$\alpha_w^{(t+1)} = \frac{\sum_{i=1}^{T} P(G_i = \Phi_w \mid \widetilde{\Theta}^{(t)})}{T}$$

$$\beta_{wk}^{(t+1)} = \frac{M\sum_{i=1}^{T}\sum_{j=1}^{U_i}\vec{a}_{ij}P(o_{ij}=\omega_{wk}\mid G_i,\Theta_w^{(t)})P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})}{\sum_{i=1}^{T}U_i}$$

$$\vec{\mu}_{wk}^{(t+1)} = \frac{\sum_{i=1}^{T}\sum_{j=1}^{U_i}\vec{a}_{ij}P(o_{ij}=\omega_{wk}\mid G_i,\Theta_w^{(t)})P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})}{\sum_{i=1}^{T}\sum_{j=1}^{U_i}P(o_{ij}=\omega_{wk}\mid G_i,\Theta_w^{(t)})P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})} \tag{7}$$

$$\Sigma_{wk}^{(t+1)} = \frac{\sum_{i=1}^{T}\sum_{j=1}^{U_i}(\vec{a}_{ij}-\vec{\mu}_{wk})(\vec{a}_{ij}-\vec{\mu}_{wk})^T P(o_{ij}=\omega_{wk}\mid G_i,\Theta_w^{(t)})P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})}{\sum_{i=1}^{T}\sum_{j=1}^{U_i}P(o_{ij}=\omega_{wk}\mid G_i,\Theta_w^{(t)})P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})}$$

$$\vec{\gamma}_{wcd}^{(t+1)} = \frac{\sum_{i=1}^{T}\sum_{\sigma=1}^{U_i}\sum_{\tau=1}^{U_i}\vec{b}_{i\sigma\tau}P(o_{i\sigma}=\omega_{wc}\mid G_i,\Theta_w^{(t)})P(o_{i\tau}=\omega_{wd}\mid G_i,\Theta_w^{(t)})P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})}{\sum_{i=1}^{T}\sum_{\sigma=1}^{U_i}\sum_{\tau=1}^{U_i}P(o_{i\sigma}=\omega_{wc}\mid G_i,\Theta_w^{(t)})P(o_{i\tau}=\omega_{wd}\mid G_i,\Theta_w^{(t)})P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})}$$

$$\Lambda_{wcd}^{(t+1)} = \frac{\sum_{i=1}^{T}\sum_{\sigma=1}^{U_i}\sum_{\tau=1}^{U_i}(\vec{b}_{i\sigma\tau}-\vec{\gamma}_{wcd})(\vec{b}_{i\sigma\tau}-\vec{\gamma}_{wcd})^T P(o_{i\sigma}=\omega_{wc}\mid G_i,\Theta_w^{(t)})P(o_{i\tau}=\omega_{wd}\mid G_i,\Theta_w^{(t)})P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})}{\sum_{i=1}^{T}\sum_{\sigma=1}^{U_i}\sum_{\tau=1}^{U_i}P(o_{i\sigma}=\omega_{wc}\mid G_i,\Theta_w^{(t)})P(o_{i\tau}=\omega_{wd}\mid G_i,\Theta_w^{(t)})P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})}$$

where

- The superscript t denotes the number of iterations.
- $P(G_i=\Phi_w\mid\widetilde{\Theta}^{(t)})$ denotes the probability of G_i matching to the model component Φ_w given the whole parameter set of the PARG model $\widetilde{\Theta}=\widetilde{\Theta}^{(t)}$.
- $P(o_{ij}=\omega_{wk}\mid G_i,\Theta_w^{(t)})$ denotes the probability of o_{ij} matching to ω_{wk} given the facts that G_i matching to the model component Φ_w and the parameter set of Φ_w is $\Theta_w=\Theta_w^{(t)}$.

4) Step (d) & (e)

Since we select some SARGs to initialize the components of the PARG model Γ, the structure of the PARG model may not be appropriate after initialization. Therefore, after each stop of the Expectation-Maximization iteration, we need to examine if the learned PARG model is a well representation of the SARGs. If the number of the model components M has been initialized too large, the SARGs will be over-represented by the learned PARG model. This can be found out by examining the corresponding α_w of the model Φ_w. If the value of α_w is too small, say smaller than a user-defined threshold, it means that the pattern represented by Φ_w only occur a few times in the SARGs. We can then delete Φ_w.

On the other hand, it is very likely to initialize the components of the PARG model with spurious nodes and relations because the SARGs may not only contain the instances of the patterns. To identify those spurious nodes, we can examine the corresponding β_{wk} of a model node ω_{wk}. If β_{wk} is too small, it means only a few sample nodes are similar to ω_{wk}. Hence, ω_{wk} should be a spurious model node and be deleted consequently. A reasonable threshold for β_{wk} could be $(1-\varepsilon)SM\alpha_w / \sum_{i=1}^{S} U_i$, where $\varepsilon < 1$.

IV. Experimental Results

This section show some examples of using the proposed approach to summarize various kinds of information.

A. Visual Information Summarization from Independent Images

We first apply the proposed approach to automatic summarizing visual information from multiple images without high-level annotation. Our approach can be directly applied to image pixels because each pixel can be treated as a node in an ARG. However, in this experiment, the resolution of the images is 307,200 pixels. Working directly on image pixels results in high computational complexity because there will be too many nodes and relations. Low-level image processing, for example, segmentation, can be used to reduce the dimension of the data and consequentially reduce the computational complexity. The MPEG-7 community has developed many tools for helping users segment and annotate images [10], [17], [24]. In this experiment, we did not use high-level annotation information because different annotator may have different understanding about the same images. We only use the low-level image features.

We segment the sample images using a segmentation program developed by Felzenszwalb and Huttenlocher [7]. The segmented images are represented as SARGs. Each node of the SARGs represents an image segment. The attribute vector of the node represents the mean and variance of the color feature (RGB) of the segment. The selected features of the segments turn out to be enough in this experiment. The adjacent relations among the segments are considered. During the learning procedure, the adjacent relations are updated as continuous variables in the range of [0, 1]. A threshold of 0.5 is used at the end of the learning procedure to decide the final values (0/1) of the relations.

An example is shown here. The sample images include 20 images of ZIP™ and 20 images of McDonald™ observed in various backgrounds and different viewpoints. Some of them are shown in Figure 7(a). Those images are segmented (see Figure 7(b)) and represented as ARGs (see Figure 7(c)). We initialize the PARG model with four model components and set the constraints as $\alpha_w > 0.5$ & $\beta_{wk} > 0.8SM\alpha_w / \sum_{i=1}^{S} U_i$. After learning, the PARG model only has two model components. One model component has six model nodes. The other one has eight model nodes. The images are also classified into two classes, which of them is represented by one of the model components. To visualize the results, we used each learned model component to detect the instances

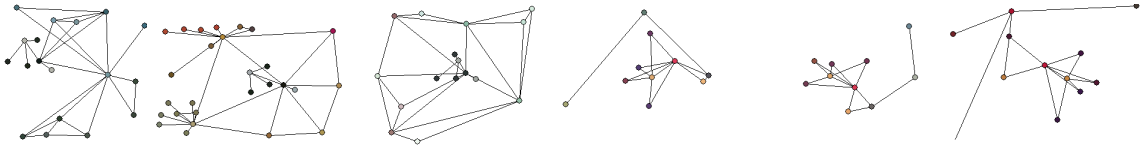
of patterns in the SARGs of its class. The detected instances of the PARG model components are shown in Figure 7(d). The original image segments corresponding to the detected pattern instances are shown in Figure 7(e).



(a) A subset of the sample images.



(b) The segmentation results of the images.



(c) The ARG representations of the images.



(d) The instances of the PARGs detected from the SARGs.



(e) The original image segments corresponding to the instances of the PARGs.

Figure 7. Visual information summarization.

B. *Summarizing Textual Information*

Consider the publication list in the following example (The list is part of the publication list of this paper's authors. The reason of using this data is for convenient purpose and to avoid legal issues.)

- Pengyu Hong, Zhen Wen, and Thomas S. Huang, "Real-time speech driven expressive synthetic talking faces using neural networks," *IEEE Transaction on Neural Networks*, January 2002.
- Pengyu Hong and Thomas S. Huang, "Unsupervised temporal pattern extraction and association," *International Conference on Acoustics, Speech, and Signal Processing*, 2002.
- Pengyu Hong, Zhen Wen, and Thomas S. Huang, "An integrated framework for face modeling, facial motion analysis and synthesis," *ACM Multimedia*, 2001.

- Pengyu Hong and Thomas S. Huang, "Learning the isomorphic sub-graph from multiple attributed relation graphs," *International Workshop on Combinatorial Image Analysis*, 2001.
- Pengyu Hong, Qi Tian, and Thomas S. Huang, "Incorporate support vector machines to content-based image retrieval with relevant feedback," *7th IEEE International Conference on Image Processing*, 2000.
- Thomas S. Huang, Ira Cohen, Pengyu. Hong, and Yang Li, "IA3: intelligent affective animated agents," *International Conference on Image and Graphic Technology toward 21 Century and Beyond*, 2000.
- Roy Wang, Pengyu Hong, and Thomas Huang, "An automatic moving object extraction system for video indexing," *International Conference on Pattern Recognition*, 2000.

We convert each item in the above list into a structural graph as Figure 8:

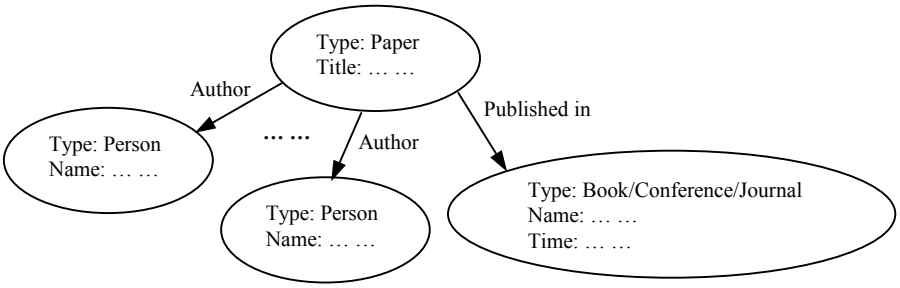


Figure 8. The ARG structure of publications. The number of author nodes can be one or more than one.

Then we apply the proposed approach to summarize the above list. The PARG model is set to have only one component. We also require each model node should have at least four instances in the ARGs of the above list. Different results are obtained if different methods are used to measure the similarity of nodes. Some results are shown in Figure 9. As shown, the summarization results heavily depend on the similarity measurements.

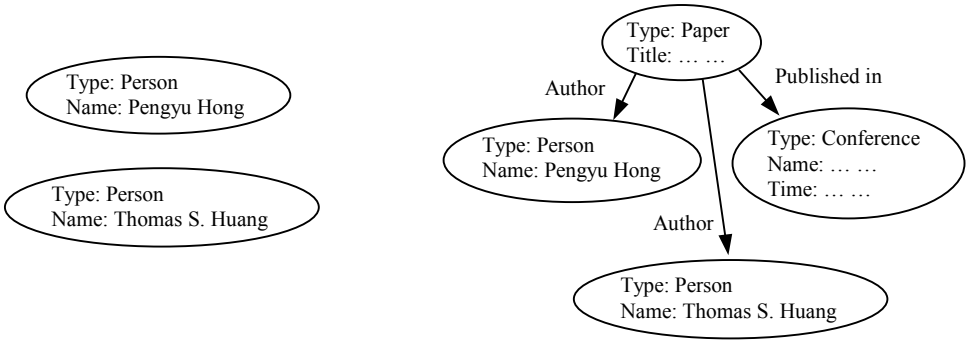


Figure 9. Summarizing results of a publication list. (a) Both of the type and the value of the attribute vector of the node are required to be the same. (b) If nodes are of type "Person", the values of their attribute vectors are required to be the same. Otherwise, only the types of nodes should be the same. Despite of similarity requirement,

C. Summarizing Data Containing Heterogeneous Information

The proposed approach can be used to summarize data containing heterogeneous information. We collected seven video clips in the same day. Those video clips describe different persons activities at different gas stations of one company. Those activities are represented by ARGs, which contain heterogeneous nodes. Some examples are shown in Figure 10.. Figure 10. (a) describes the video clip of a person adding gas to his car at the gas station. The video clip of Figure 10. (b) shows that a person stopped at the gas station and was calling somebody. The video clip of Figure 10. (c) shows that person stopped at the gas station to buy drink. Each gas station is associated with a representative picture.

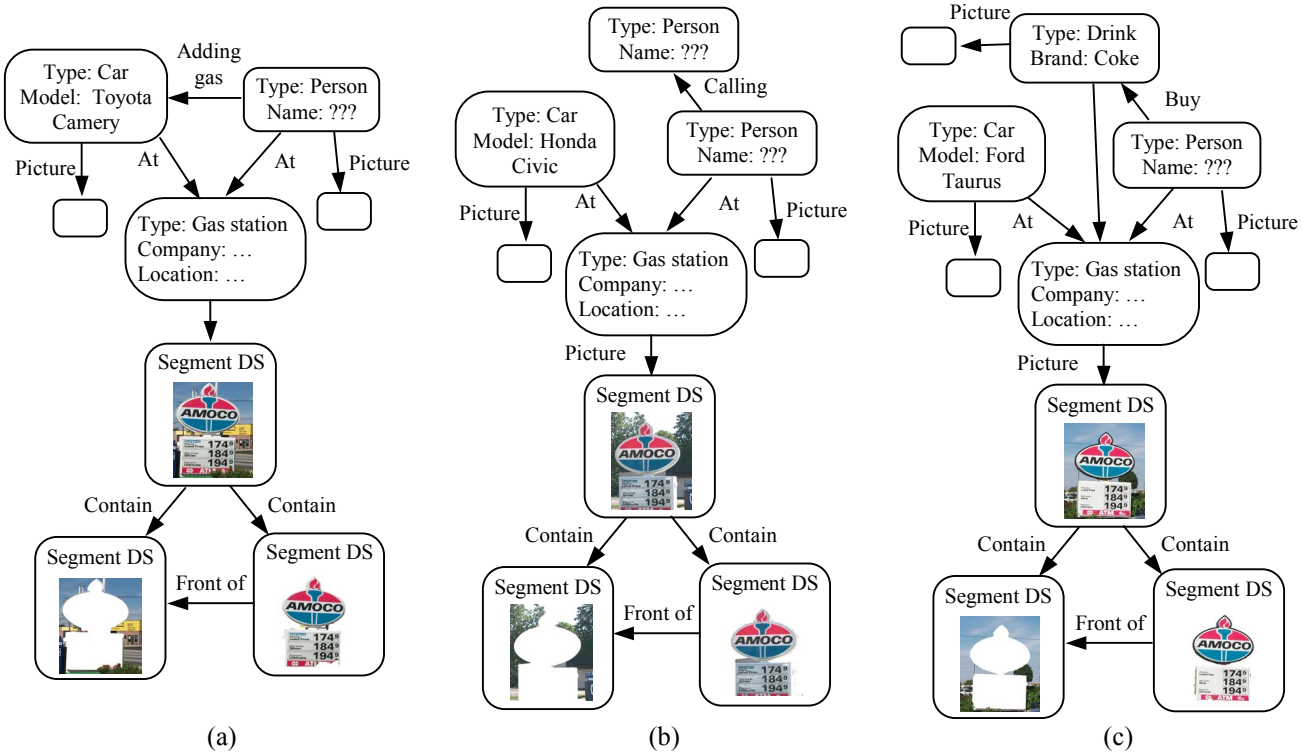


Figure 10. The ARGs of the activities at gas stations. The information about persons is missing because we don't know who they are. Some information is not shown (e.g., the pictures of cars and persons, etc.)

As shown in last experiment, we know that the summarization results depend on the similarity measurement. The following similarity measurement criteria are used by the summarization procedure.

(a) If the type of a node is "person", "car", only the type similarity is checked because we are more interested in what happened to those persons and their cars.

(b) If the type of a node is "Gas station", we only check the similarities of type and "Company" attribute.

(c) There are two similarity measurements for those nodes whose type is "Segment DS" of the MPEG-7. If a "Segment DS" node is a parent of other "Segment DS" node, we only check type similarity because it is difficult to tell what is in the image if the features of the objects in the image are mixed together. If a "Segment DS" node has no "Segment

DS" nodes as children, the similarities of low-level features (color histogram and texture histogram) are checked.

(d) A mode node should have at least 4 instances in the ARGs of seven video clips.

The PARG model is set to have only one component. The summarization results are shown as the instances of the PARG model in Figure 11.

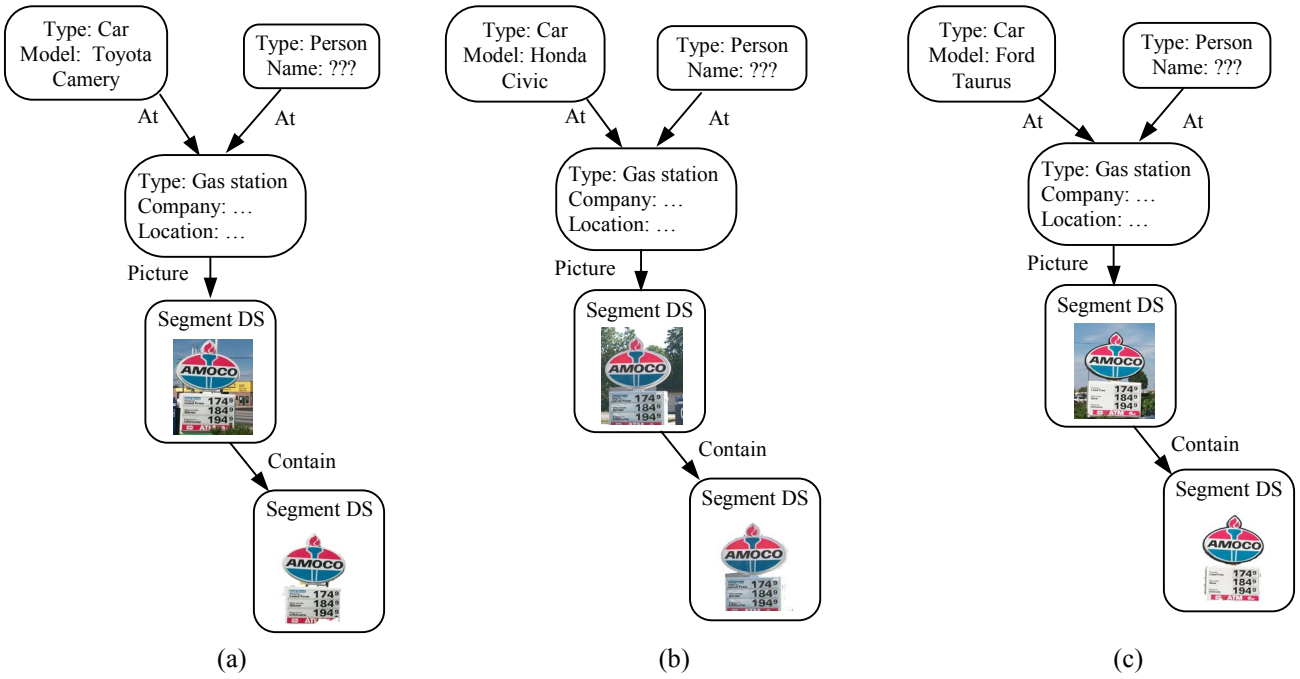


Figure 11. The summarization results are shown as the instances of the PARG model in the examples shown in **Figure 10**.

From the above summarization results, we cannot make a conclusion about the main activities of different persons at the gas stations. This may due to we don't have enough observations. However, the summarization results include image segments describing the gas prices and the logo of the company, which is very difficult to describe in text. This example shows that the combination of heterogeneous information may provide rich and efficient description of information.

V. CONCLUSIONS

This paper presents the methodology for information retrieval and summarization in the context of the MPEG-7 standard and the Semantic Web. Based on the study of the data representation of both the MPEG-7 standard and the Semantic Web, we find that both communities can use a unified representation – ARG – to represent information. We then point out the problems of information retrieval and summarization in the context of the MPEG-7 standard and the Semantic Web are equal to graph matching and summarization. Algorithms are presented for graph retrieval and summarization. Experimental results are used to demon-

strate the effectiveness of the proposed methodology.

Some problems are opened for future investigation. Graph matching techniques play an important role. It is interesting to investigate how to impose all kinds of constraints (e.g., hierarchical constraint) brought in by the MPEG-7 standard and the Semantic Web initiative on the calculation procedure of graph matching. The proposed summarization algorithm has high computational complexity. Given the daily explosively increasing amount of multimedia information, online efficient summarization techniques are highly demanded, however challenging.

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